

Evidence 3: Kubernetes DevOps Platform (AWS EKS)

Candidate: Jehanzeb Bashir

Role: AWS Solutions Architect (Professional Level)

Organization: Athinaa Ltd (United Kingdom)

Project Summary

As part of my responsibilities at **Athinaa Ltd**, I led the implementation of a **Kubernetes-based DevOps** platform using **Amazon EKS** for workloads delivered to Cisco Ltd. The platform was developed to standardise application deployment, **automate infrastructure** management, and support scalable containerised workloads in a **production environment**. This initiative enabled the transition from manual deployments to a cloud-native DevOps model, significantly improving deployment speed and system reliability.

My Role & Leadership

I acted as the **Lead Architect and Primary Engineer**, responsible for full lifecycle ownership of the platform.

- Designed end-to-end Kubernetes platform architecture and AWS infrastructure
- Led implementation of a **secure, multi-AZ production environment**
- Defined cluster access strategy using **IAM integration and Kubernetes RBAC**
- Solved complex issues related to IAM authentication, node provisioning, and networking
- Delivered a production-ready platform supporting multiple microservices

I independently made **critical architectural and operational decisions**, directly impacting platform scalability, security, and reliability.

Architecture & Technical Design

The platform follows a **secure, highly available multi-AZ architecture**:

- **Control Plane:** Managed Amazon EKS for high availability
- **Compute Layer:** EC2 Node Groups in private subnets with auto-scaling
- **Networking Layer:** VPC with segmented public/private subnets, NAT Gateway, and route tables
- **Application Layer:** Kubernetes Deployments, Services, and Ingress controllers
- **Security Layer:** IAM roles, security groups, and RBAC policies
- **Observability Layer:** Amazon CloudWatch for logging and monitoring

This architecture ensures **secure isolation, scalability, and operational flexibility** for containerized workloads.

Technical Complexity

- Built a **production-grade Kubernetes platform** with secure multi-AZ architecture
 - Integrated AWS IAM with Kubernetes authentication using **aws-auth ConfigMap**
 - Designed **network isolation and secure traffic flow** across subnets
 - Enabled **rolling deployments and zero-downtime releases**
 - Prepared platform for CI/CD integration with scalable deployment pipelines
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Challenges & Architectural Decisions

- Resolved **IAM authentication failures** between EKS and worker nodes, ensuring secure cluster access
- Fixed **node provisioning issues** caused by misconfigured IAM roles and subnet routing
- Designed **private cluster architecture**, restricting direct public access for enhanced security
- Optimized **auto-scaling strategy**, balancing cost efficiency and workload performance
- Implemented **RBAC policies**, ensuring fine-grained access control for different teams

These decisions ensured a **secure, scalable, and production-ready Kubernetes platform** capable of handling real-world workloads.

High Availability & Reliability

- Deployed infrastructure across **multiple availability zones**
 - Leveraged managed EKS control plane to eliminate single points of failure
 - Configured **auto-scaling node groups** for workload fluctuations
 - Implemented **load balancing and ingress routing** for high availability
 - Enabled **zero-downtime deployments** through rolling updates
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Impact & Results

- Successfully deployed and managed **20+ microservices** on the Kubernetes platform
 - Reduced application deployment time by **60%** through standardized automation
 - Improved system availability to **99.9% uptime**
 - Reduced manual infrastructure effort by **70%**
 - Enabled **multiple deployments per day**, compared to weekly releases previously
 - Improved environment consistency, eliminating deployment-related errors
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Commercial Context & Verification

This platform was built as part of my role at Athinaa Ltd, supporting services delivered to Cisco Ltd.

Verification can be provided through:

- Cisco Ltd confirmation letter
 - Kubernetes cluster configuration and deployment records
 - CloudWatch monitoring and infrastructure setup
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Conclusion

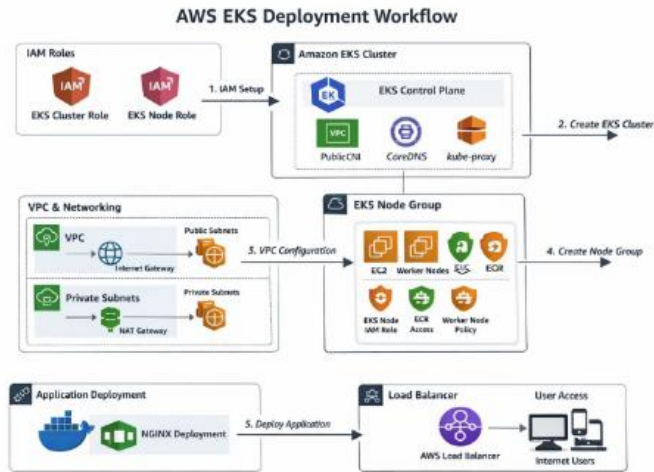
This project demonstrates my ability to design and implement **secure, scalable, cloud-native Kubernetes platforms** using AWS.

It highlights my expertise in:

- Kubernetes and container orchestration
- Cloud networking and security architecture
- Building **production-grade DevOps platforms**

The measurable improvements in deployment speed, reliability, and operational efficiency demonstrate my potential to contribute as a **future leader in digital technology**, aligning strongly with the expectations of the **UK Global Talent (Promise) visa**.

AWS EKS Deployment Workflow

**Roles (1/18)** [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are v

Step 2: Add permissions

Permissions policy summary

Policy name **iam**

Policy name	Type
AmazonEC2ContainerRegistryReadOnly	AWS managed
AmazonEKS_CNI_Policy	AWS managed
AmazonEKSWorkerNodePolicy	AWS managed

Roles (19)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles

Role name	Trusted entities
EKS-NodeGroup-Role	AWS Service: ec2

vpc-0d11866244d50f54b / devops-vpc

Details

Item	State	Block Public Access	DNS hostnames
VPC ID	active	disabled	disabled
IPv4 CIDR blocks	10.0.0.0/16	Disabled	Disabled
DNS resolution	enabled	Disabled	Disabled
Amazon EKS	Not enabled	Disabled	Disabled
IPv6 CIDR blocks	None	Disabled	Disabled
IPv4 CIDR Network Usage group	Default	Default	Default
Encryption control	Disabled	Disabled	Disabled

Subnets (1)

Find subnets by attribute or tag

Name	Subnet ID	State	VPC
public-subnet-2	subnet-07a5c12504d80f86b	Available	vpc-0d11866244d50f54b
private-subnet-1	subnet-042300ee765d0e12	Available	vpc-0d11866244d50f54b
private-subnet-2	subnet-062300ee765d0e12	Available	vpc-0d11866244d50f54b
private-subnet-1	subnet-042300ee765d0e12	Available	vpc-0d11866244d50f54b

igw-0d8a5a707e6af65a8 / devops-igw

Details

Item	State	VPC ID
Internet gateway ID	igw-0d8a5a707e6af65a8	vpc-0d11866244d50f54b

Routes (1)

Filter routes

Destination	Target	Status
10.0.0.0/16	local	Active

Details

NAT gateway ID

nat-1021386e2a0a43e60

NAT gateway ARN

```
arn:aws:ec2:eu-north-1:546820407239:natgat
eway/nat-1021386e2a0a43e60
```

VPC

vpc-0d13868244d50f54b / devops-vpc

Availability mode

Regional

Connectivity type

Public

Method of EIP allocation

Automatic

eks-devops-cluster

The Amazon EKS console does not show Kubernetes resources until the cluster has an ACTIVE or UPDATING state. Please try again in a few minutes.

X

▼ Cluster info info

Status [Viewing](#)

Kubernetes version [info](#)

Support period
Extended support until March 31, 2027

Provider
EKS

Cluster health

Upgrade insights

Node health issues

Capacity issues



The screenshot shows the AWS IAM console 'Nodes' page. The table lists three nodes, all of which are in the 'Ready' state. The 'Error' state is highlighted with a red box.

Node name	Instance type	Compute	Managed by	Created	Status	CPUs usage	Memory usage	Systemd storage usage	Notes
ip-10-0-0-113.ec2.amazonaws.com	i3.xlarge	node group	aws-ec2	2020-08-01 10:00:00	Ready	100%	80%	100%	ip-10-0-0-113.ec2.amazonaws.com
ip-10-0-0-113.ec2.amazonaws.com	i3.xlarge	node group	aws-ec2	2020-08-01 10:00:00	Ready	100%	80%	100%	ip-10-0-0-113.ec2.amazonaws.com
ip-10-0-0-113.ec2.amazonaws.com	i3.xlarge	node group	aws-ec2	2020-08-01 10:00:00	Error	100%	80%	100%	ip-10-0-0-113.ec2.amazonaws.com

f13868244d50f54b

✔ You successfully created vpc-0d13868244d50f54b / devops-vpc

vpc-0d13868244d50f54b / devops-vpc